

CE2007

Microprocessor-based System Design

The Analogue Interface

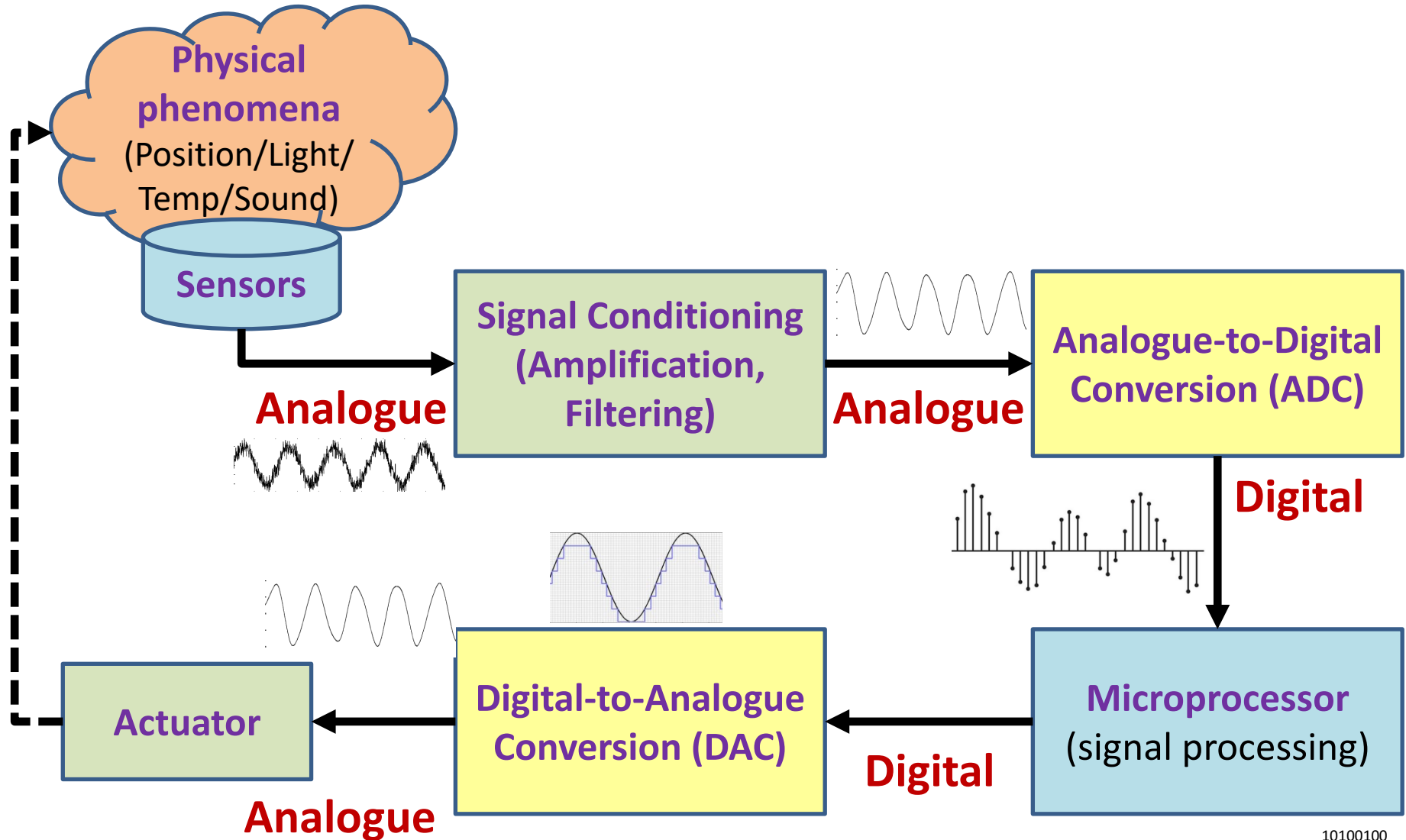
Topics Covered

- Signal Acquisition
- Digital-to-Analogue Conversion (DAC)
- Analogue to Digital Conversion (ADC)

Reference: Alan Clements, “Principles of Computer Hardware”, 4th Edition, Oxford University Press (Section 11.6)

Signal Acquisition

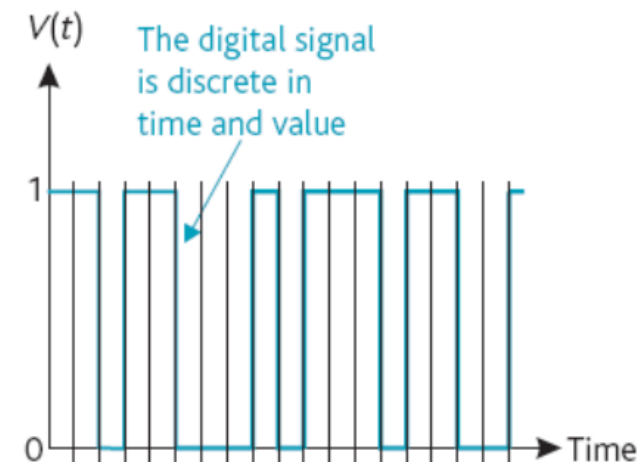
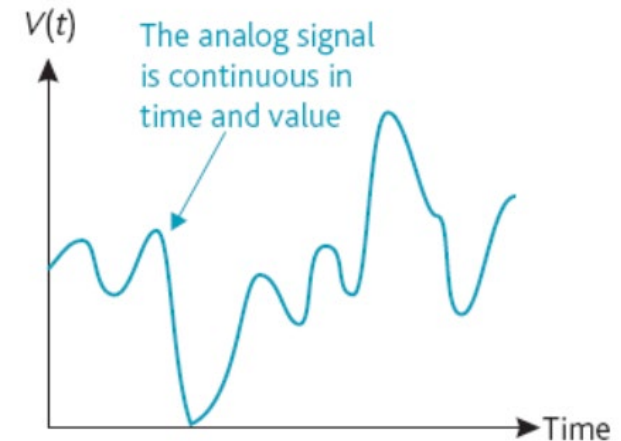
The Measurement and Control Loop



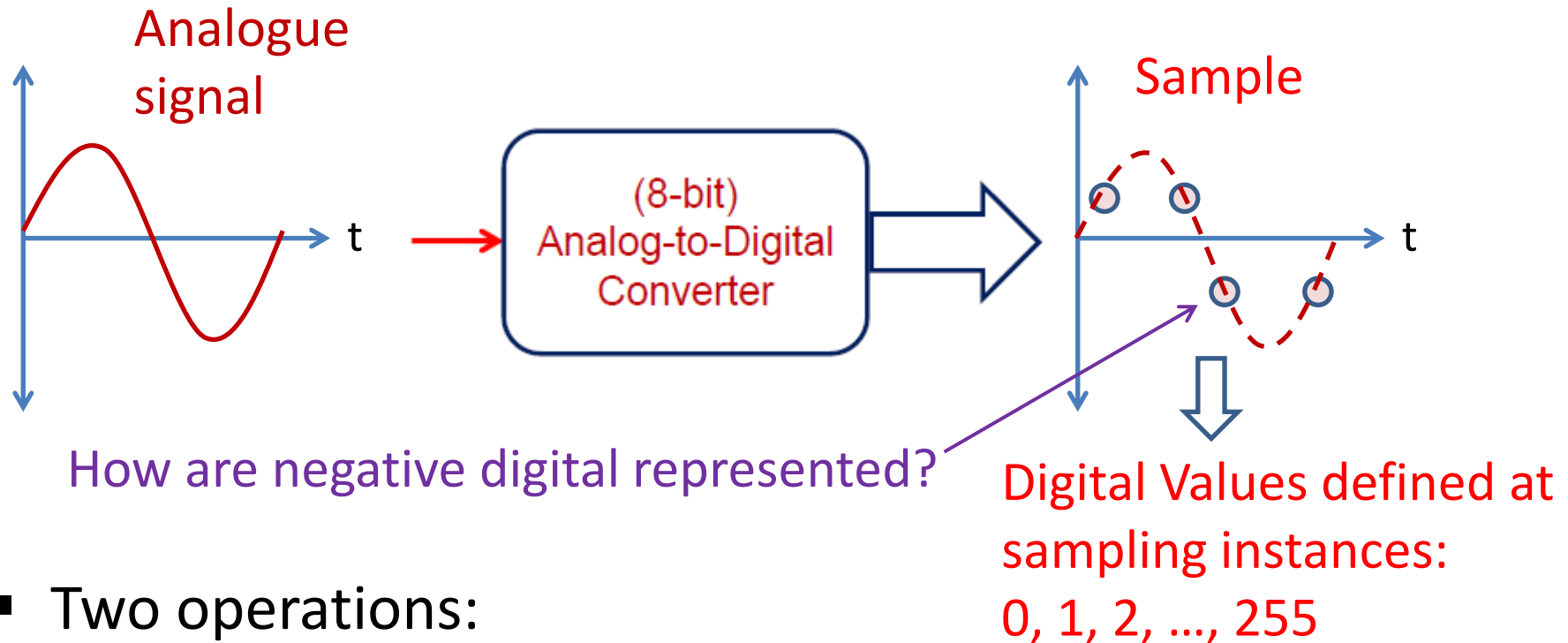
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Analogue vs. Digital Signals

- When microprocessor is used to interact with real world signals (which are mainly analog in nature),
 - Converters such as Analog-to-Digital Converter (**ADC**) and Digital-to-Analog Converter (**DAC**) are needed
- Analog Signal
 - Continuous in time and amplitude
 - i.e. **infinite resolution**
- Digital Signal
 - Signal valid at discrete time only (due to sampling)
 - Amplitude valid at certain discrete levels only
 - i.e. **finite amplitude and time resolution**



Analogue to Digital Conversion

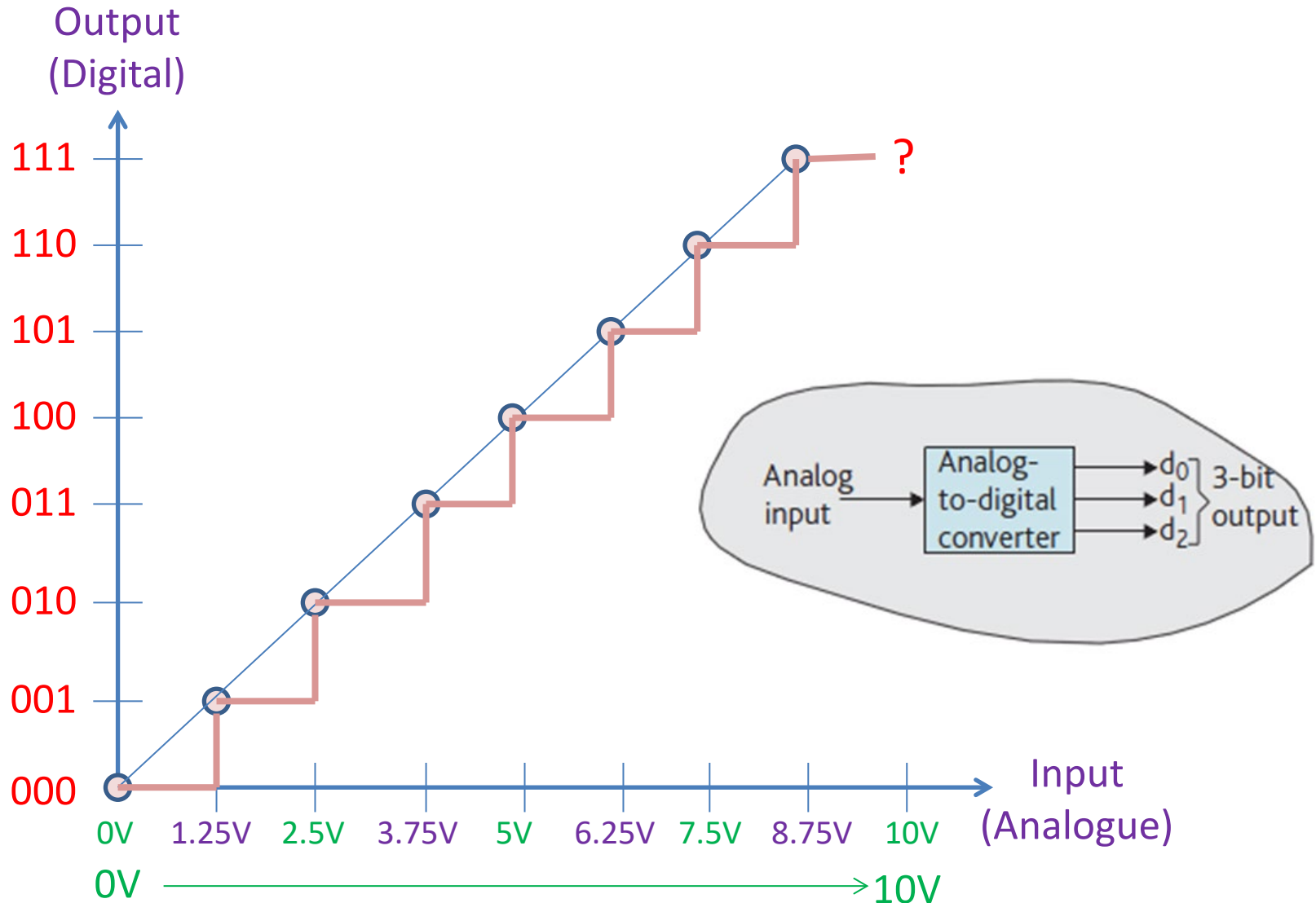


- Two operations:
 - Extract sample value of analogue signal
 - Convert sample value to digital value (in binary)
- Hence, there is loss of signal's detail when it is converted from analog to digital

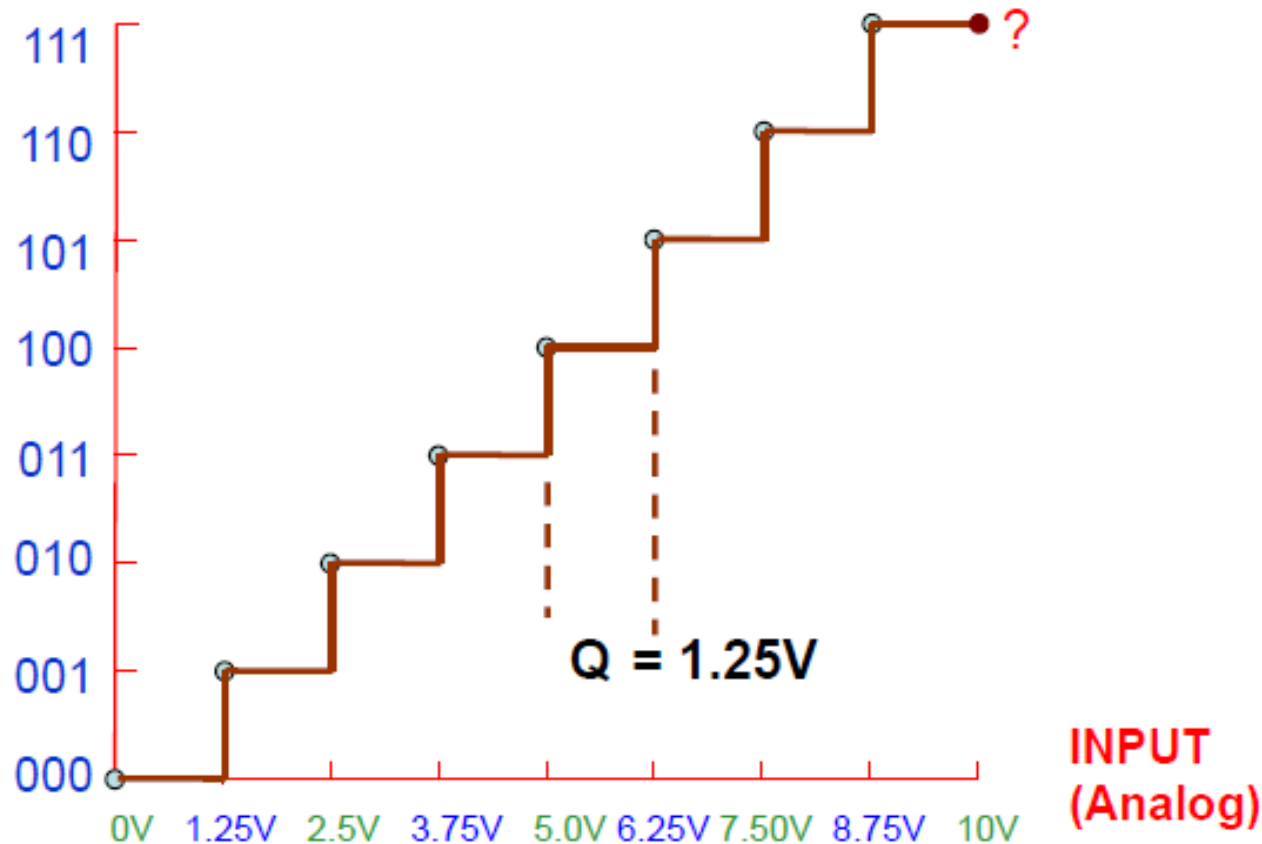
- Analogue to digital conversion requires **quantization** of continuous signal into multilevel signals
 - Results in loss of signal resolution
- Two fundamental questions:
 - How often should the conversion (sampling) is carried out?
 - How many levels (digital values) should the analogue input be divided? For example,
 - Input voltage range is 0V to 10V (can have any values in this range e.g. 1 mV, 1 μ V, etc.) – Infinite analogue values
 - Output values of a 3-bit ADC is only 0, 1, 2, .., 7 (8 digital values)

Input-Output Characteristics of an ADC

Example: A 3-bit ADC Input-Output Characteristics



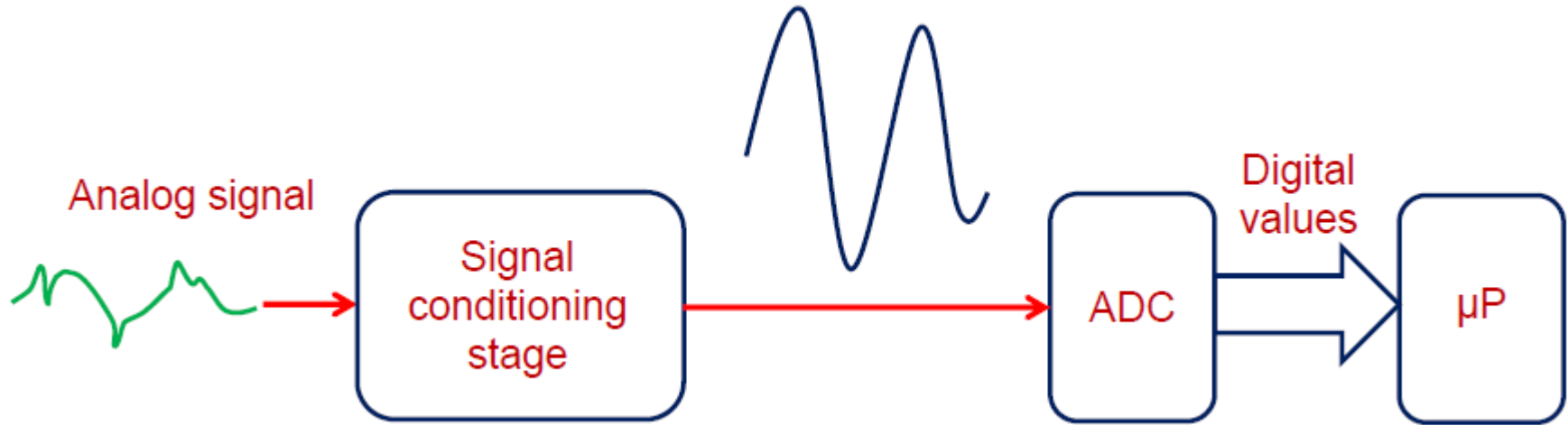
Quantization Q



- Quantization Size (Quantum) $Q = FS/2^N = 10/2^3 = 1.25V$
 - The input voltage range for which its digital output value remains the same

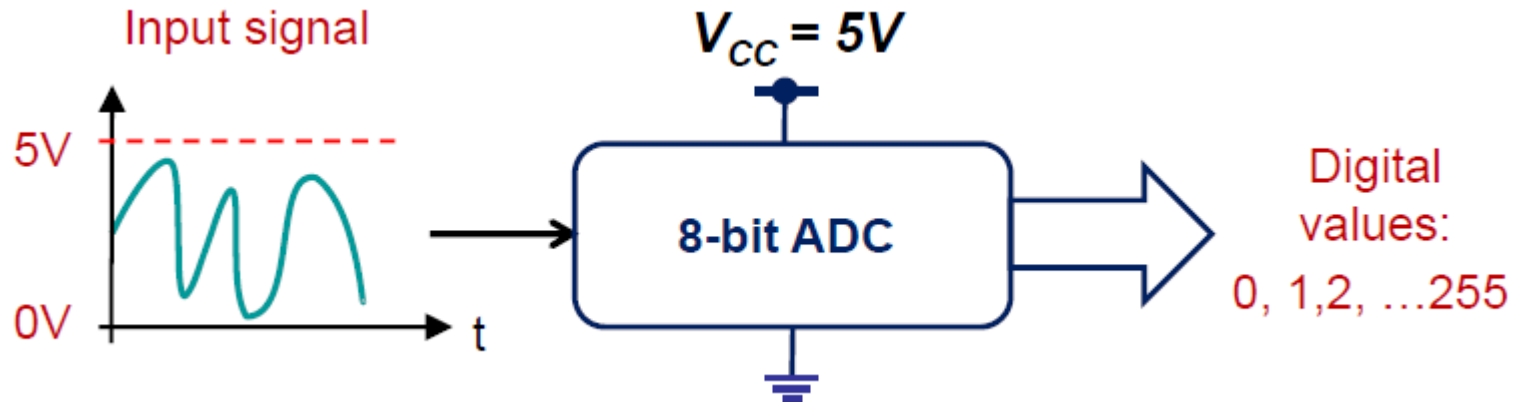
- Resolution of ADC can be expressed/described in term of
 - Number of bits (N) of ADC (e.g. 8-bit, 10-bit)
 - Actual Quantization size $Q = FS/2^N$
 - Where FS = Full Scale input
 - e.g. 1.25V (for a 0V-10V, 3-bit ADC)
 - 2.44mV (for a $\pm 5V$, 12-bit ADC)
- Since $FS/2^N = \text{Least Significant Bit size}$
 - Q is also described as 1 LSB

Input Signal Requirement



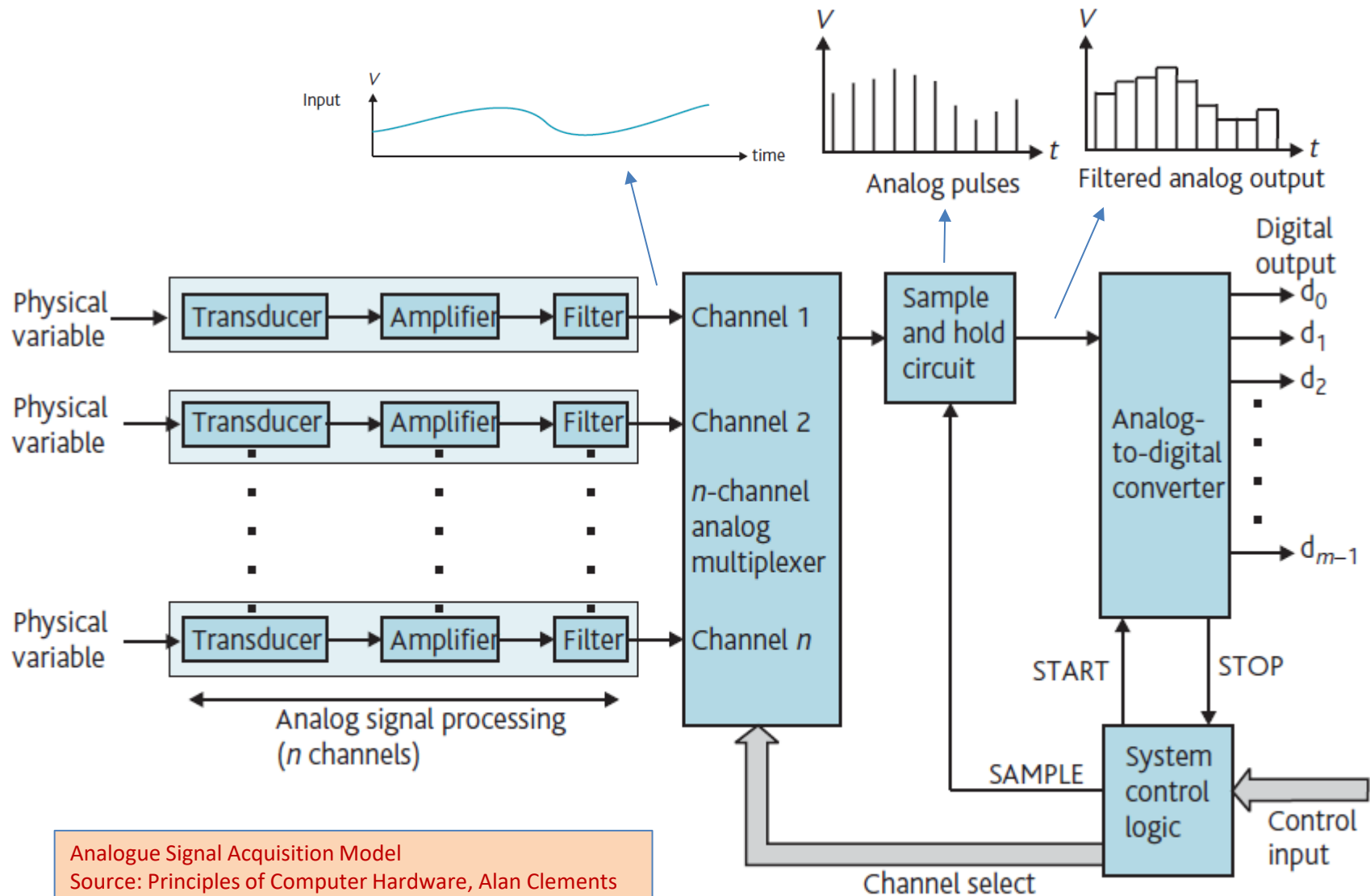
- What should be the signal characteristic at the input to the ADC (i.e. output of the signal conditioning stage)?
 - Amplitude should have a maximum level close to the Full Scale (FS) input to provide highest digital resolution
 - Noise level should be less than the quantization size (Q) or LSB such that it doesn't affect the signal value

Example



- Power supply of ADC $\approx$ FS input of the ADC
 - Hence, max input signal $\leq 5V$
- Quantization size (Q) of 8-bit ADC $= 5V/256 = 19.5mV$
 - Hence, noise at input must be $< 19.5mV$

An Analogue Signal Acquisition Model



Sampling and Aliasing

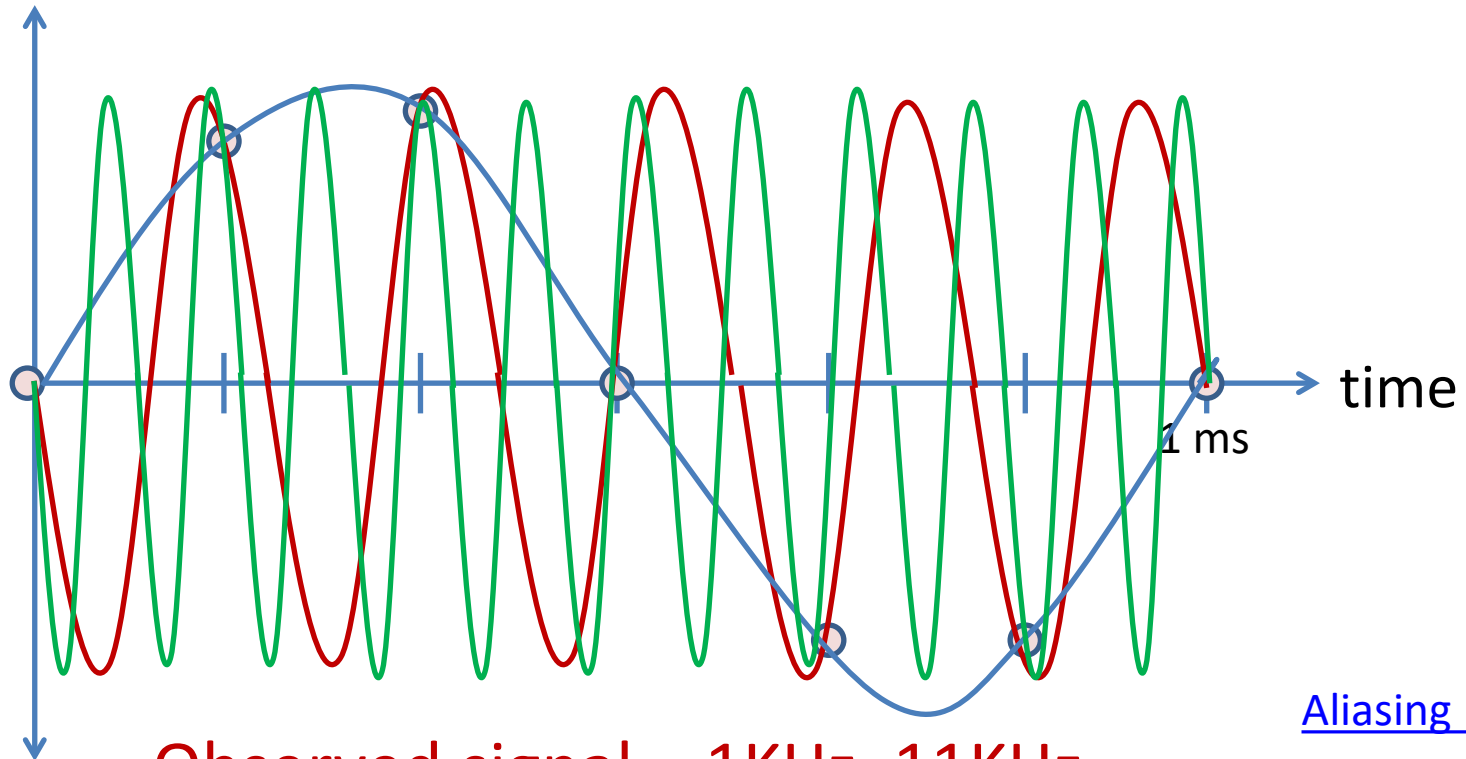
- Another important consideration when applying ADC is the sampling frequency used in the system
 - i.e. the number of samples of the signal taken over one second duration
- Intuitively, the more frequently we sample the signal
 - The more accurate is the digital values we obtained
- Correspondingly, if the number of samples taken is too little
 - The less precise is the information we captured
- In Analogue to Digital conversion, the sampling frequency must be **more than two times** of the highest frequency component in the signal (Nyquist rate)
 - otherwise **aliasing** effect will occur and information is lost

Aliasing

- Reconstruction from acquired signal samples introduces distortion if sampling rate is smaller than Nyquist rate

Signal = 5KHz (5 cycles in 1ms)

Sampling frequency = 6KHz (6 samples in 1 ms)



Observed signal = 1KHz, 11KHz,

[Aliasing Example](#)

DAC and ADC

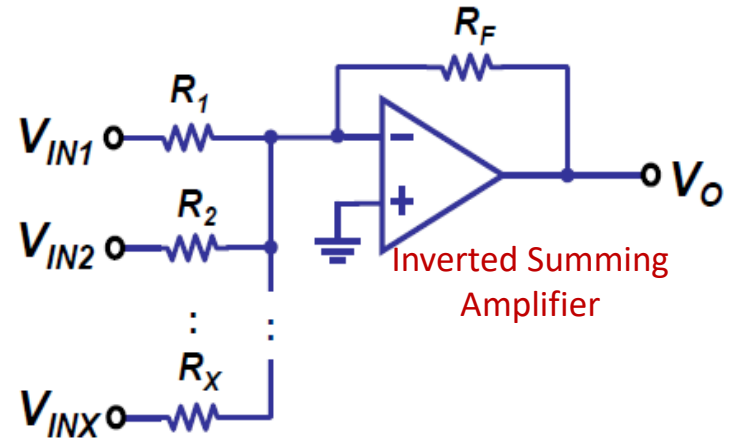
Digital to Analogue Conversion

- Real world signals are more easily processed, stored and transferred in digital format
 - but need to eventually converted back to human recognizable analogue form
 - converted using Digital-to-Analogue converter (DAC)
- Examples:
 - Generation of audio signals in music players
 - MP3 files
 - Compact Disc
 - Generation of video signals in video player
 - DVD player
 - Video files replay on smart-phone/tablet

DAC Building Blocks

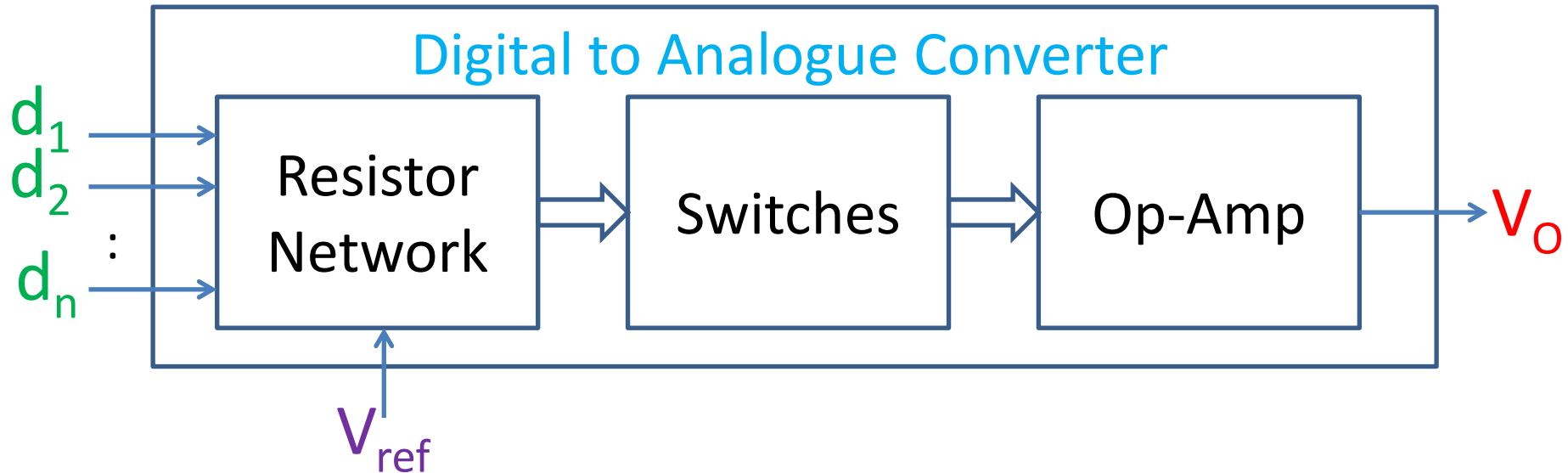
- DAC converts a binary value to analogue form
- This is achieved by:

- Generating an analogue value that is proportional to each bit of the digital word
- Adding the analogue values to give a composite analogue sum



$$V_O \approx K \bullet D$$

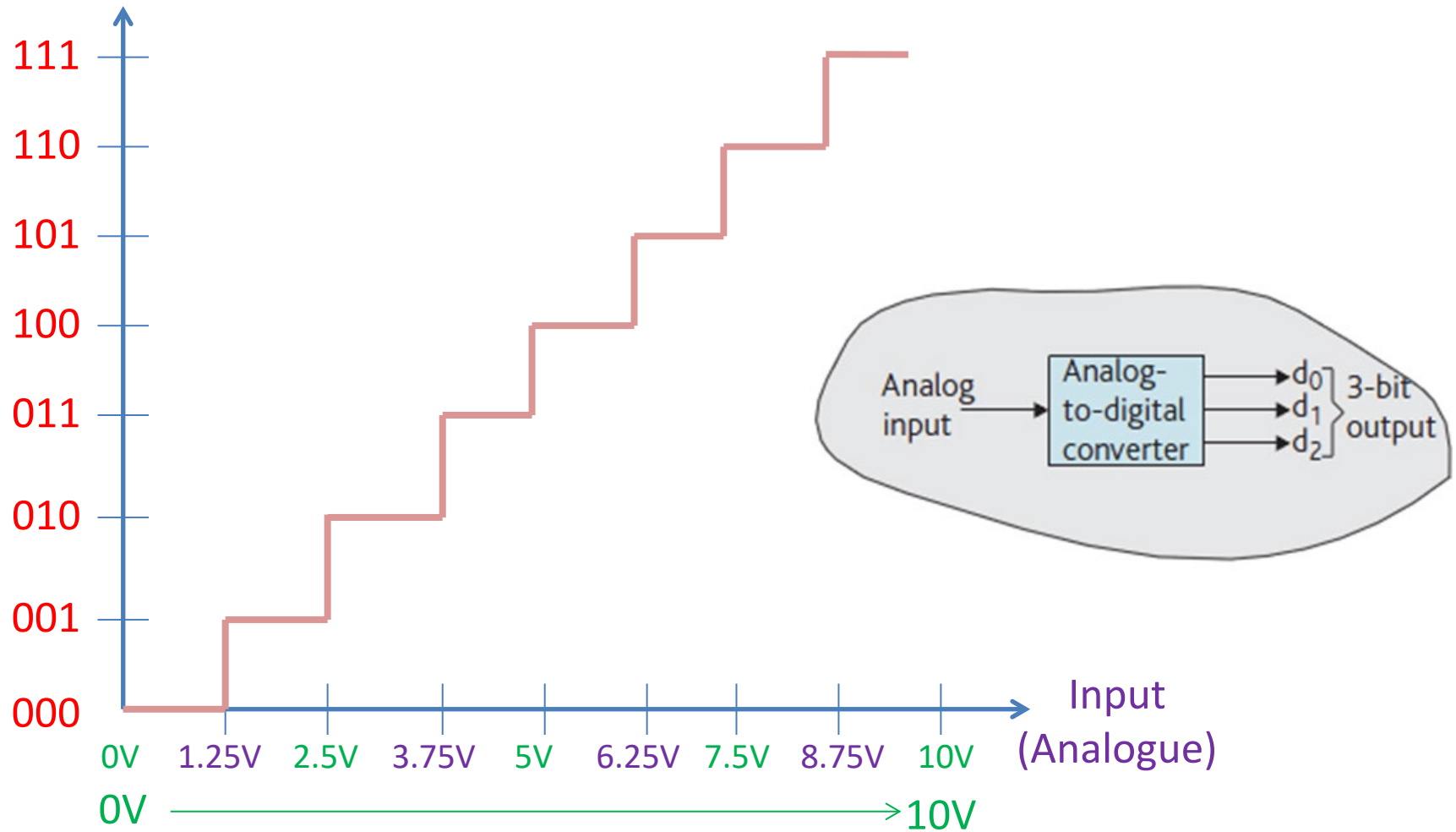
$$D = d_1 2^{-1} + d_2 2^{-2} + \dots + d_n 2^{-n}$$



Types of ADC

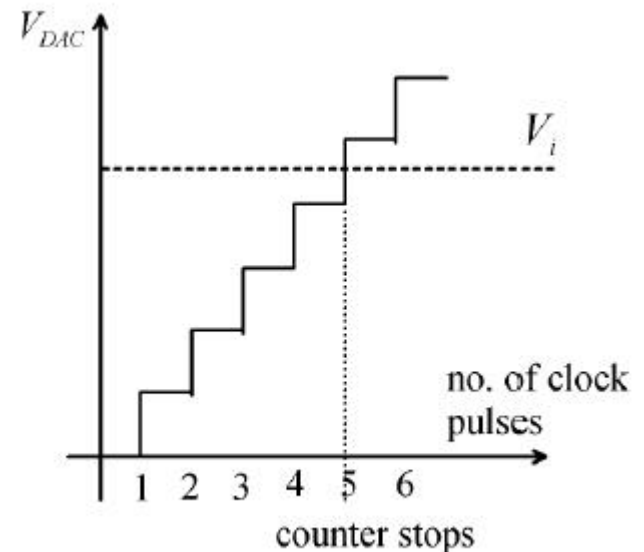
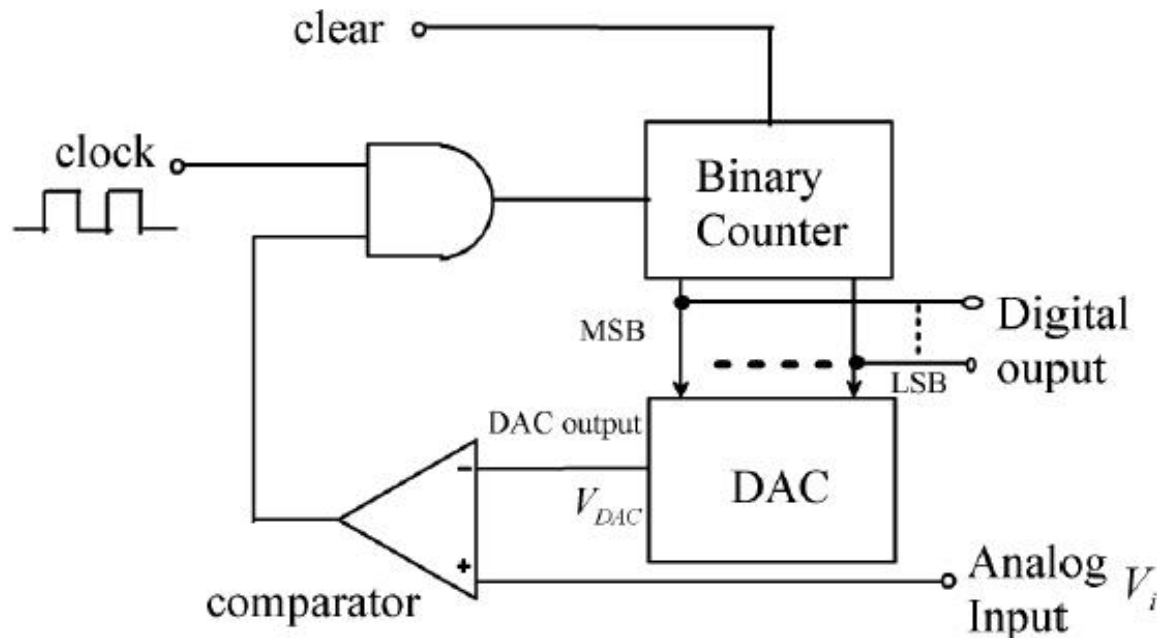
- Some types of ADC available in the market
 - Flash (Parallel)
 - Digital Ramp
 - Tracking
 - Successive Approximation
 - Integrating
 - Sigma-Delta
- Each type operates based on different principle
 - Different performance parameters (e.g. conversion time)
 - Cost effectiveness
 - Different application requirements

Input-Output Characteristic of an ADC

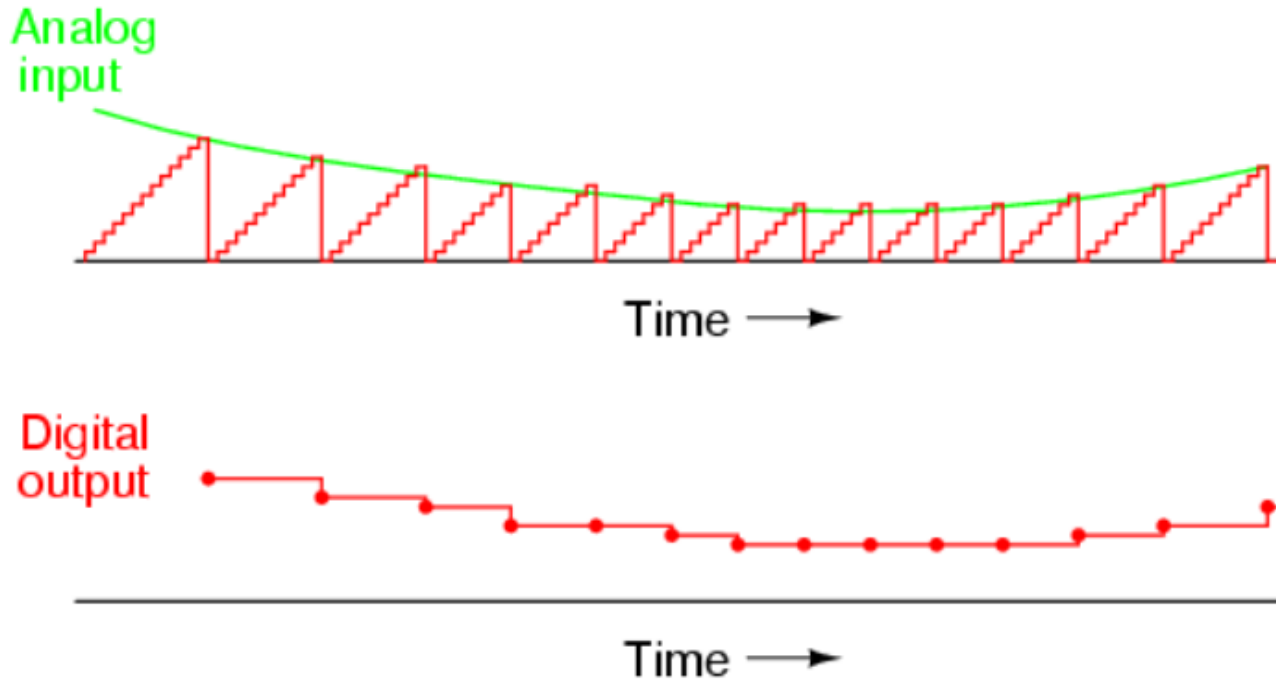


Digital Ramp ADC

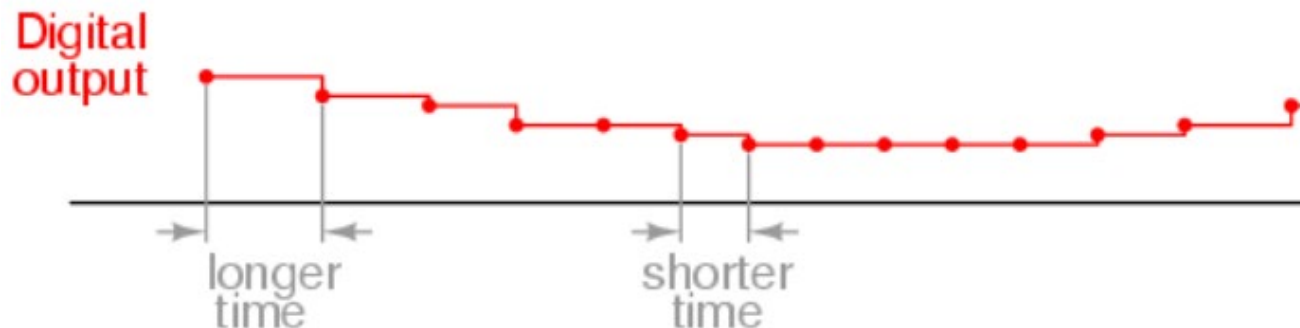
- Connect the output of a free-running binary counter to the input of a DAC
- Compare the analogue output of the DAC with the analogue input signal V_{in}
- Use the comparator's output to tell the counter when to stop counting and reset



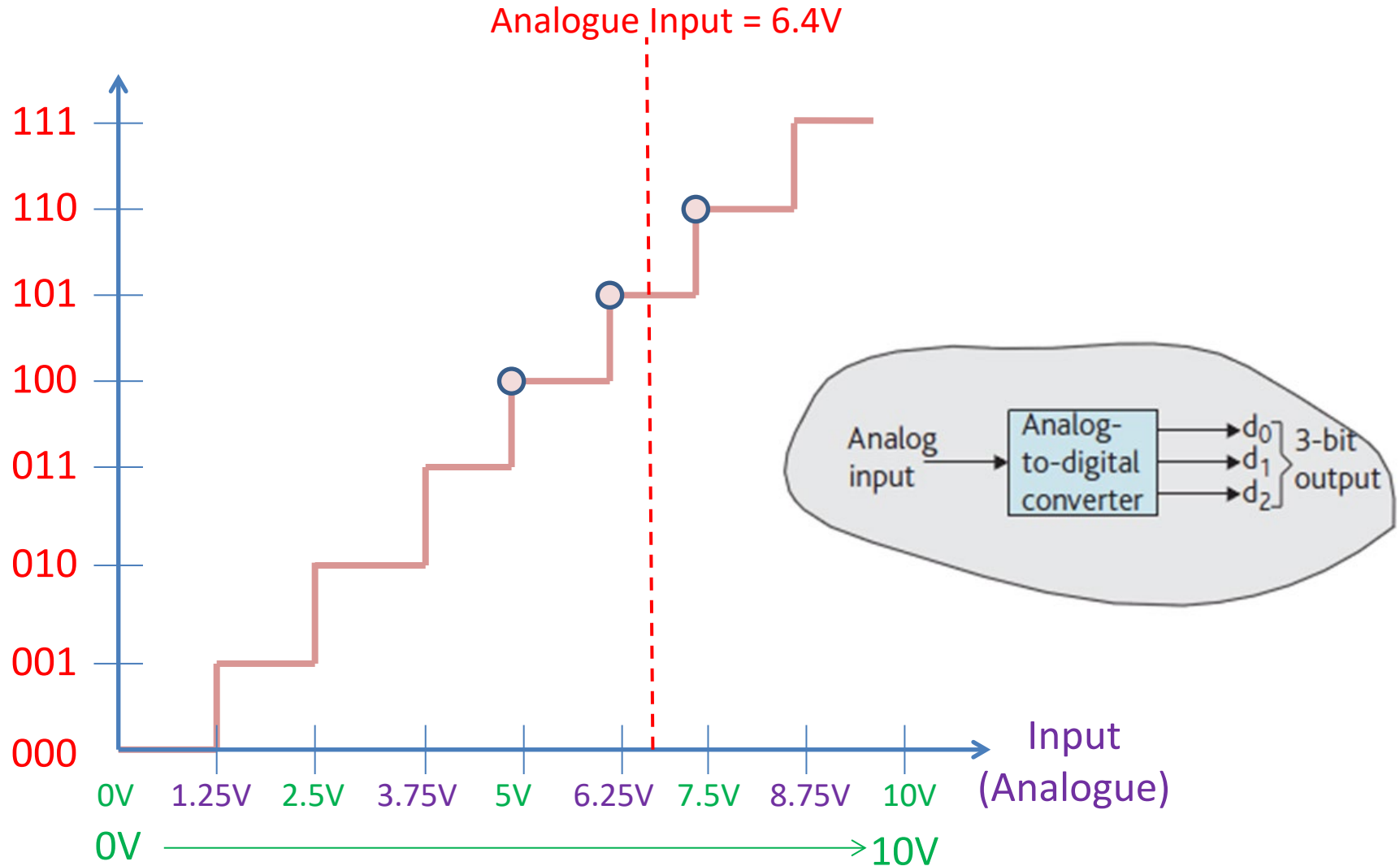
Digital Ramp ADC (Cont.)



- **Disadvantage:** Time between new digital output values changes depending on how high the input voltage is

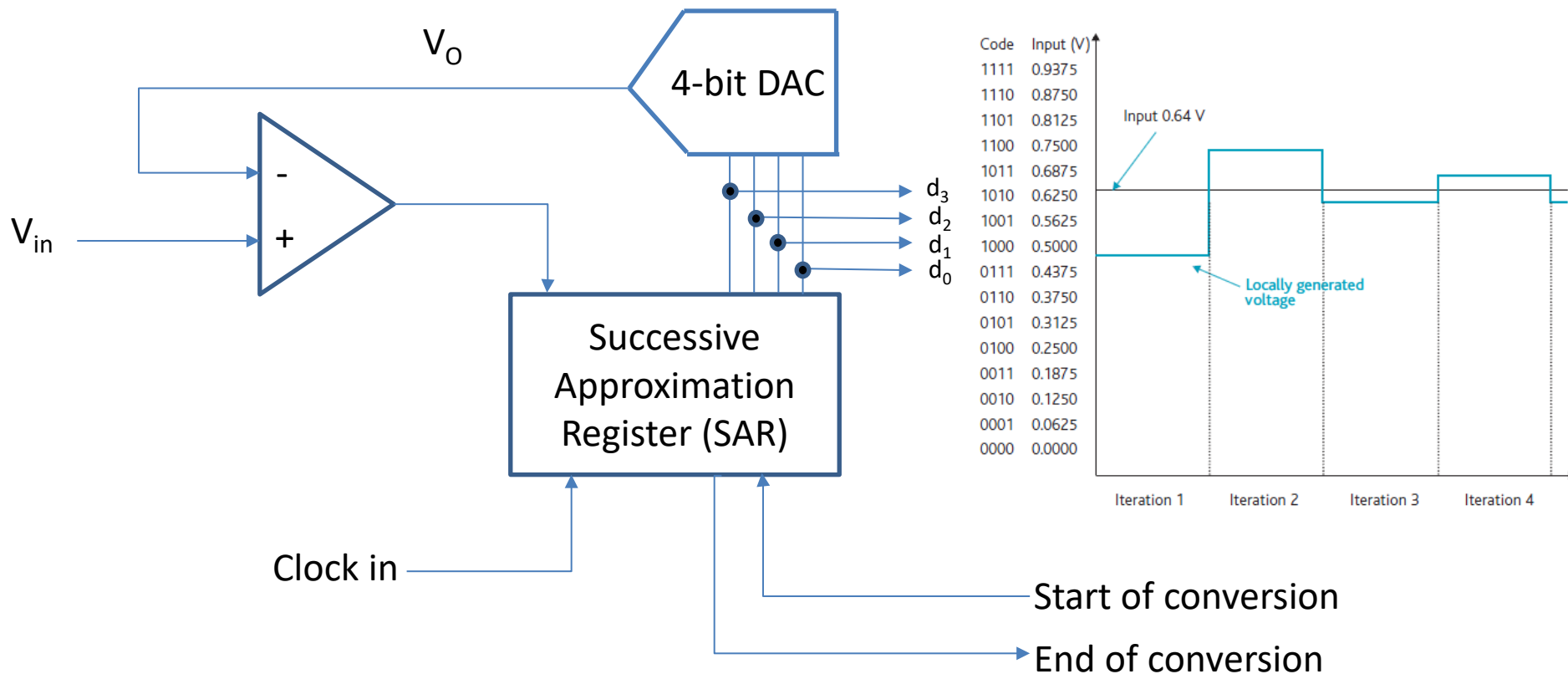


Input-Output Characteristic of an ADC



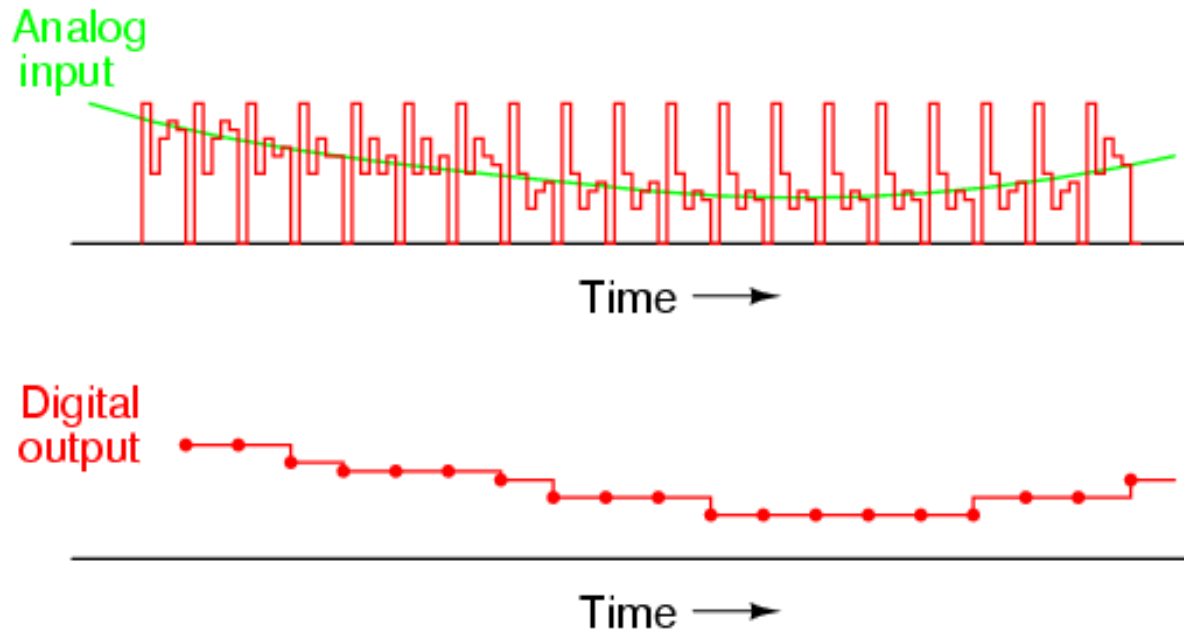
Successive Approximation

- Idea is to **take large steps** in increasing the locally generated analog signal (V_o) in the beginning, and reduce the step size as V_o converges to V_{in}



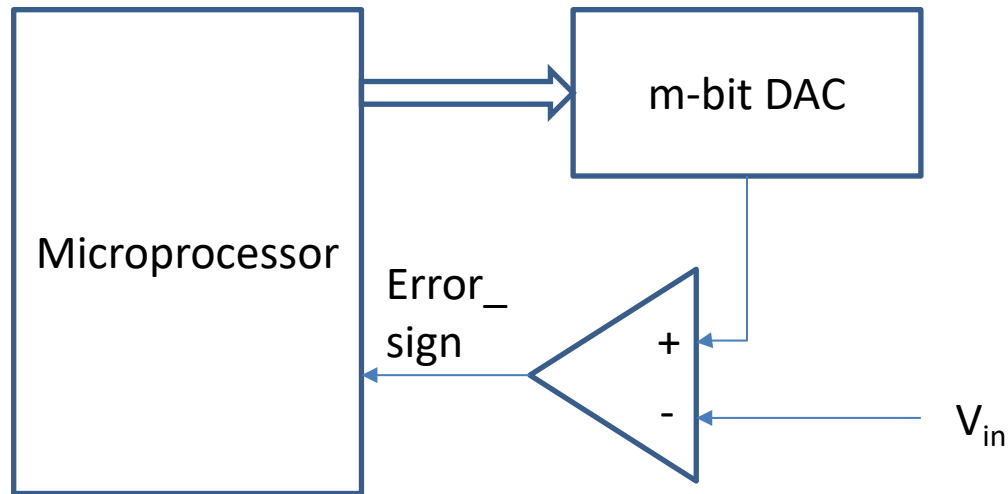
Successive Approximation (Cont.)

- Guarantees an **m-bit conversion** in **no more than m iterations**
- Produces faster results than the digital ramp ADC



- Can be implemented partially in software to reduce hardware cost

Successive Approximation (Cont.)



Successive Approximation

```
DAC_output = 0
Increment  = ½ Full-Scale Output (1000 ... 000)
For I = 1 to m
    DAC_output = DAC_output + Increment
    If Error_sign negative then
        DAC_output = DAC_output - Increment
    End If
    Increment = Increment/2
End For
```

End Successive Approximation